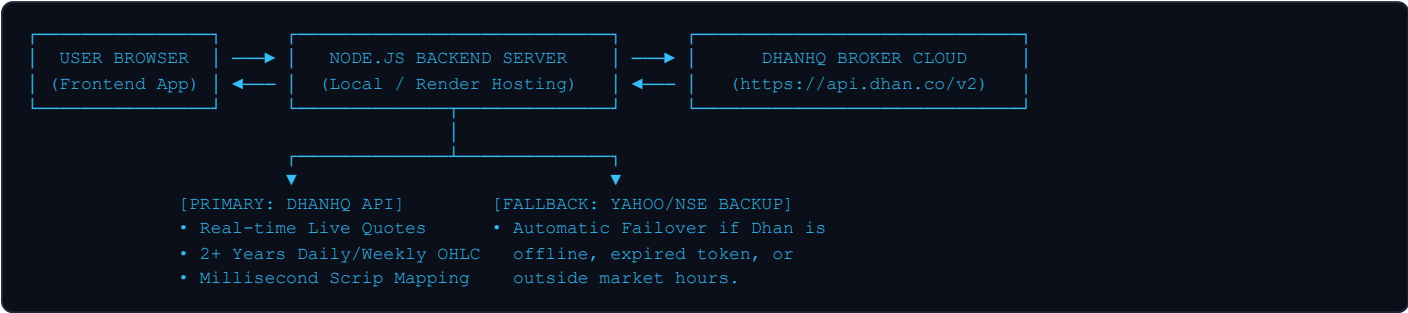


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Complete technical blueprint for DhanHQ Live Broker API & Historical Engine

SECTION 1 : DHAN API ARCHITECTURE

1. DhanHQ End-to-End Request Pipeline Live Market Data & Historical Candles Flow



2. How Dhan API Connection Works (Step-by-Step for Freshers) Core Mechanism

🔑 Step 1: Authentication & Credentials

Our server reads two environment variables from .env:

- **DHAN_CLIENT_ID**: Your unique 10-digit Dhan account number.
- **DHAN_ACCESS_TOKEN**: Secure 256-bit JWT access token generated in the Dhan developer portal.

⚡ Step 2: Scrip Master Security Mapping

Dhan identifies stocks by numeric IDs, not names. Our server bundles dhan_scrip_map.json (10,000+ securities) to instantly translate symbols:

"RELIANCE" → SecurityID: 2885 (NSE_EQ)
"NIFTY 50" → SecurityID: 13 (IDX_I)

📊 Step 3: Historical Candle Conversion

Calls POST https://api.dhan.co/v2/charts/historical. Dhan returns raw epoch array data. Our server converts it to TradingView OHLCV format: { time: "2026-09-04", open, high, low, close, volume }

🛡️ Step 4: Graceful Auto-Fallback Engine

If Dhan token expires, returns error, or is unconfigured, the app **never crashes**. It seamlessly switches to Yahoo Finance/NSE backup with zero user interruption.

3. Dhan API Endpoints & Payloads Reference API Specifications

FEATURE	ENDPOINT URL & METHOD	REQUEST HEADERS	PAYLOAD STRUCTURE
Historical Charts	POST /charts/historical	client-id, access-token	{ securityId: "2885", exchangeSegment: "NSE_EQ", instrument: "EQUITY", fromDate, toDate }
Live Quotes (LTP)	POST /marketfeed/ltp	client-id, access-token	{ "NSE_EQ": [2885, 1594, 3045, 11536] } (Batch up to 100 stocks in 1 request)

Live Feed Status: Look at the top-right navbar badge: ● **Dhan** indicates active live broker stream, while **Backup** indicates automatic fallback feed.

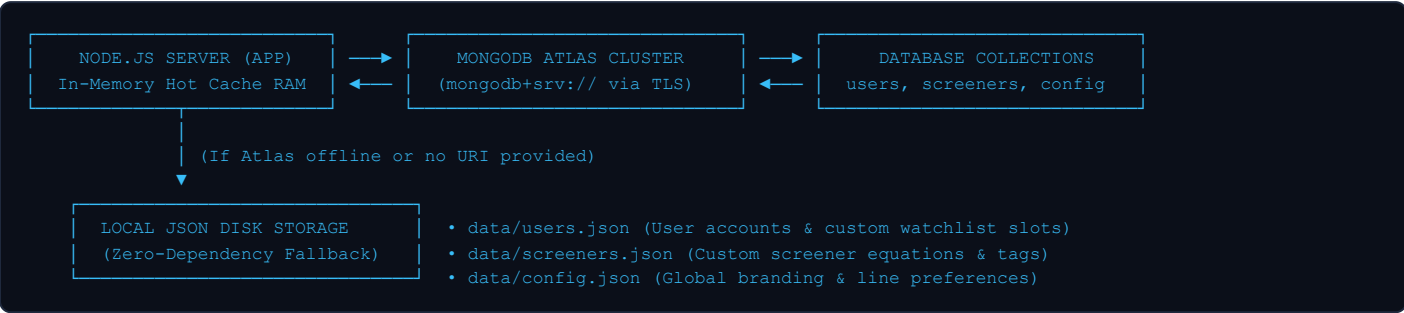
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Complete technical blueprint for MongoDB Atlas Cloud Database & Local JSON Fallback

SECTION 2 : MONGODB ATLAS ARCHITECTURE

1. MongoDB Atlas Dual-Storage Topology

Cloud Database & Local JSON Sync



2. How MongoDB Connection Works (Step-by-Step for Freshers)

Connection Lifecycle

Step 1: DNS SRV Lookup Resolution

Atlas uses mongodb+srv:// URLs. Node.js resolves SRV records into 3 distributed replica-set server hostnames using configured DNS resolvers (8.8.8.8, 1.1.1.1).

Step 2: TLS/SSL Encrypted Handshake

A secure SSL socket connects to the Atlas cluster. Authentication verifies the database username, password, and IP access list (0.0.0.0/0 for cloud hosts).

Step 3: Smart Auto-Seeding & Ingest

On first connect, if Atlas collections are empty, the server automatically seeds default screeners and admin accounts directly into Atlas with zero setup needed.

Step 4: Dual-Write & Local Backup

Every time you create a screener or save a watchlist, the server writes asynchronously to MongoDB Atlas **AND** local JSON files, guaranteeing zero data loss.

3. MongoDB Collections Schema & Purpose

Data Model

COLLECTION NAME	DOCUMENT SCHEMA KEYS	USAGE & FUNCTIONALITY
users	userId, username, passwordHash, role, watchlists[5], preferences	Multi-user accounts, salted passwords, user-specific watchlists (5 x 50 stocks), indicator line widths and color preferences.
screeners	id, name, description, chartinkUrl, category, tags, updatedAt	Custom screeners, Chartink query URLs, category classification, and scanner logic.
config	systemName, maxUserSlots, globalThemes, lastMaintenance	Global system configuration, user registration capacity slots, and maintenance flags.

4. Common Troubleshooting for Freshers

Error Handling

COMMON ISSUE	ROOT CAUSE	INSTANT FIX
Invalid Scheme Error	Accidental spaces in .env or pasting MONGODB_URI=mongodb+srv://... in value	Trim URI string. Server auto-strips duplicate prefixes now.
SSL Alert Number 80	Node TLS strict certificate or DNS SRV timeout	Auto-handled by dns.setServers() and dual-mode JSON fallback.
IP Not Whitelisted	MongoDB Atlas Network Access blocking server IP	In MongoDB Atlas → Network Access → Add IP 0.0.0.0/0 (Allow anywhere).